

Serial LCD Module



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In most electronic systems, it is necessary to provide information in the visual form for a convenient user interface. The most commonly used displays in electronic systems are LED, LCD and TFT. Amongst these, an LCD display is the most cost-effective solution as it can display alphanumeric characters along with special user-defined symbols. It also contains an integrated controller which frees up the main microprocessor from the task



Fig. 1: Author's prototype

of refreshing the display and improves overall system throughput. But at the same time it requires a large number of pins for interfacing with the system, which consume a lot of microcontroller's ports.

The presented serial LCD module can convert any 16x2 alphanumeric LCD display to a serial LCD which communicates with the main controller using serial communication technique. The brightness of LCD backlight can also be controlled. If the text exceeds 16 characters, the remaining text automatically accommodates in the second line. The serial LCD module communicates on 9600 standard baud rate, which is compatible with all microcontrollers. Fig. 1 shows the author's prototype of the serial LCD module.

Circuit and working

Fig. 2 shows the circuit of the serial LCD module connected to Arduino Uno (Board1) using only one wire. Microcontroller ATmega8 (IC1) works

with an 8MHz internal oscillator. It uses no-xtal bootloader which provides Arduino compatibility without 16MHz external crystal. Using 8MHz internal oscillator reduces hardware and improves power efficiency. IC1 uses its serial port to communicate with the

TABLE I
Connections between
IC1 and LCD1

IC1 pins	LCD pins
15	15 (LED brightness control)
16	D7
17	D6
18	D5
19	D4
26	RS
25	R/W
24	EN

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	+5V (with respect to TP0)
TP2	Serial data

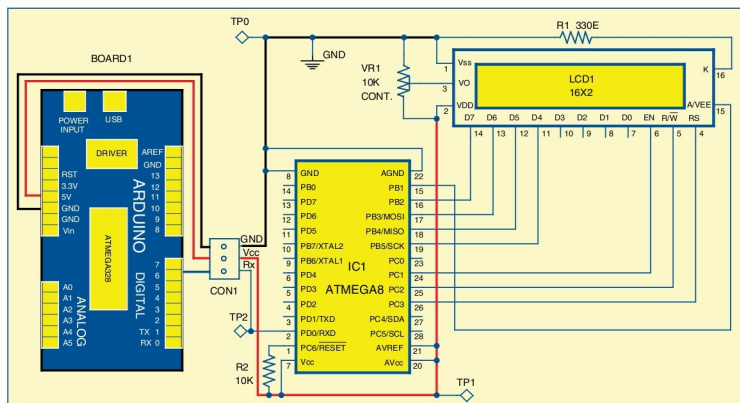


Fig. 2: Circuit diagram of serial LCD module